
Spontaneous Pattern Formation and Genetic Diversity in Habitats with Irregular Geographical Features

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Abstract: *The role of spontaneous pattern formation, the appearance of inhomogeneities that are not directly imposed by external forces, has not been closely examined in the context of the origin and maintenance of genetic diversity in wild populations. Using individual-based computer simulations, we demonstrated that such patterns form in spatially distributed species with local demes under disruptive selection. In our model systems, spatial patterns of genetic diversity arose and changed over time even in the context of a spatially homogenous environment. The spatial distribution and dynamics of the fittest genotypes were controlled by the movement of boundaries between domains of the different genotypes. The rate of diversity decay was dramatically slower than predicted by traditional models. Therefore, spontaneous pattern formation may lead to the maintenance of genetic diversity of a species in a contiguous habitat, despite reproductive mixing. Moreover, the diversity persisted significantly longer in larger habitats and habitats with irregular geographical features. Habitat structure was intimately linked to the preservation of genetic diversity. Spontaneous pattern formation should be considered along with other spatial effects in the design of conservation areas.*

Formación Espontánea de Patrones y Diversidad Genética en Hábitats con Características Geográficas Irregulares

Resumen: *El papel de la formación espontánea de patrones, la apariencia de inhomogeneidades que no son impuestas directamente por fuerzas externas, no se ha examinado detalladamente en el contexto del origen y mantenimiento de la diversidad genética en poblaciones silvestres. Utilizando simulaciones de computadora basadas en individuos, demostramos que tales patrones se forman en especies distribuidas espacialmente con demes locales bajo selección disruptiva. En nuestros sistemas de modelo, los patrones espaciales de diversidad genética surgieron y cambiaron a lo largo del tiempo aún en el contexto de un ambiente espacialmente homogéneo. La distribución espacial y la dinámica de los genotipos más aptos fueron controlados por el movimiento de límites entre los dominios de los genotipos diferentes. La tasa de descomposición de diversidad fue dramáticamente más lenta de lo predicho por modelos tradicionales. Por lo tanto, la formación espontánea de patrones puede conducir al mantenimiento de la diversidad genética de una especie en un hábitat contiguo, a pesar de la mezcla reproductiva. Más aún, la diversidad persistió significativamente por más tiempo en hábitats más extensos y en hábitats con características geográficas irregulares. La estructura del hábitat estuvo íntimamente ligada a la preservación de la diversidad genética. La formación del patrón espontáneo debe ser considerada, junto con otros efectos espaciales, en el diseño de áreas de conservación.*

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Introduction

Spontaneous pattern formation describes a process that gives rise to inhomogeneous structure within a system by internal dynamics and without external forces that impose either the existence or the shape of the structure. Five decades ago British mathematician Alan Turing presented a theoretical model of spatially extended chemical reactions that could create static patterns from homogeneous initial conditions (Turing 1952). After a long period of dormancy, the dynamics of such spatial pattern formation has recently been attracting attention in developmental biology for the purpose of describing various general phenomena such as spontaneous cell differentiation in embryos, and specific phenomena such as the formation of pigment patterns on animal skins. Besides Turing's static patterns, there are several other distinct classes of spontaneous pattern formation that theoretically are found to be generic to large classes of models of interacting systems and are realized in a wide range of experimental physicochemical systems.

Dynamic domain formation and coarsening behavior, which is reported for models of ecological systems in this article, was first recognized in statistical physics for systems in which local components take either of two (or more) stable states and local interactions encourage local homogeneity (Lifshitz & Slyozov 1961). This behavior can be seen in many natural and artificial systems, including quenched magnetic materials in which each magnetic atom becomes magnetized in one of two opposite directions, and other diverse physical systems at a phase transition (e.g., between liquid and gas or liquid and solid). The spatial dynamics and characteristics of this behavior have been extensively studied through analytical and computational methods (Bray 1994). The central feature of this behavior is the characteristic time dependence of the typical domain size that increases as a function of time (coarsening) as a result of interdomain boundary motion.

Spontaneous pattern formation has recently been introduced into ecology to study spatial interspecific interactions (Durrett & Levin 1994; Tilman & Kareiva 1997), where the competition between two or more species may create a spatial pattern of species that changes over time. However, these studies do not consider the possibility or implications of such pattern dynamics for intraspecific genetic diversity. Here we consider a theoretical model of spatially distributed mating populations with multilocus genetics and show via individual-based computer simulations that spontaneously forming spatial genetic patterns increase the persistence of genetic diversity when selection favors two fit genotypes over their genetic intermediates (disruptive selection) (Bar-Yam 1999; Sayama et al. 2000).

Moreover, we show that these patterns interact with habitat structure so that it becomes directly relevant to

the maintenance of genetic diversity. Diversity persists longer in larger habitats because of the time required for pattern coarsening by the motion of boundaries between extreme genotypes (hybrid zones). The internal structure of the habitat is also important. Localized obstacles that do not preclude interbreeding nevertheless contribute to the maintenance of genetic diversity by inhibiting the movement of hybrid zones. We found that the most effective obstacle size for maintaining diversity was, approximately, the radius of a local deme. These results are relevant to spatial ecology and conservation biology, in which both genetic diversity and the size and shape of population reserves are of concern because of their implications for the long-term survival of species (Diamond 1975; Lynch 1996; Margules & Pressey 2000).

Model

We considered a theoretical model of a spatially structured population of monoecious organisms with local but overlapping demes in which the sexual reproduction of the organisms was described by an individual-based model with multilocus genetics. An organism's genome had n loci, each of which had one of m possible alleles. One mature organism exclusively occupied one spatial site, reflecting the local carrying capacity. Organism reproduction generated P offspring at each site. Each of the offspring genomes was determined by random recombination of the genomes of two parent organisms randomly chosen within a distance R , the radius of a local deme. After reproduction, the fittest individual in the offspring produced at each site was selected to become a parent at the next generation.

This mating process may be regarded as a natural implementation of the classic isolation-by-distance models in population genetics (Wright 1969). Unlike the more usual statistical treatments, individual-based models like ours specify genetic information for each individual explicitly. Genetic crossing and natural selection are realized directly as a mating of pairs of organisms and a selection of specific organisms rather than operations applied to genetic distributions. The effectiveness of such individual-based models for addressing diverse questions has been increasingly recognized because of the simplicity of incorporating various complex situations that are less intuitive in the case of purely analytical or statistical models (Deangelis & Gross 1992).

The phenotype of each offspring was assumed to be a quantitative trait determined from its genome in such a way that each allele independently and additively contributed to the trait. Selection was applied by a fitness function that was maximal for extreme values of the quantitative trait. This has the effect of disruptive selection by selecting against organisms with intermediate trait values. It is equivalent to concave fitness sets, such

as arise when a population evolves to exploit highly disparate resource types (Thoday 1972). Such disruptive selection is considered a common and general cause of trait divergence (Dieckmann & Doebeli 1999; Kondrashov & Kondrashov 1999). The details of the model and the specific parameter settings chosen for the simulations are given in Appendix 1.

Unlike many other studies in conservation genetics that focus on the spatial patterns of neutral markers, our study focuses on dynamic changes of adaptive genetic variation within populations under disruptive selective pressure against quantitative traits. In such settings we can characterize adaptive genetic variation by trait variation because they are strongly coupled, and in what follows we used trait variance as a measure of genetic diversity.

We did not include mutation in our results. We have also carried out simulations including mutation and found that the results were not affected within the range of biologically realistic mutation rates (e.g., mutation rate per locus $\mu < 10^{-3}$). Even for unreasonably high mutation rates ($10^{-3} < \mu < 10^{-1}$), the results were only weakly affected by the slowed motion of hybrid zones. This illustrates the generic nature of coarsening behavior, which is robust to many model variations that respect the locality of interactions.

Spontaneous Pattern Formation

There was a dramatic contrast between theoretical predictions based on two different assumptions: (1) a panmictic population in which R is infinite, meaning that mating is possible between all organisms (Fig. 1a) and (2) a population with local demes in which R is significantly smaller than the radius of the habitat (Fig. 1b). In the former case the population became restricted to one type after only a few generations. Under the latter condition the organisms locally tended to assume one or the other of two fittest types. Globally, spatial patterns with domains of two different genotypes formed. Because the organisms were not physically isolated from one another, the hybrid zones between genotypes moved. The hybrid zones moved in a direction so as to reduce their local curvature (i.e., domains with a convex [bulging] shape shrank to the benefit of domains with a concave [dented] shape) (Fig. 1bi). Inclusions of one type surrounded by the other type disappeared over time. The rate of motion increased as the local curvature increased (i.e., the more sharply domain interfaces curved, the faster they moved). The spatial dynamics of the domains seen in our model were equivalent to the process of pattern formation and coarsening that occurs in nonbiological systems (Bray 1994). The robustness of these dynamics was apparent when we studied the effects of variations in the model parameters n , m , R , or P . As long

as there were two fittest genotypes under disruptive selection, a similar pattern formation and coarsening always resulted.

As mentioned above, such pattern forming and coarsening behavior has recently been proposed to occur in spatially distributed populations as a result of interspecific ecological interaction (Durrett & Levin 1994; Gandhi et al. 1998) and microevolution (Sayama et al. 2000). The key property common to both is the frequency-dependent dynamics in the local population growth, which is not captured by other purely random genetic-drift models. The spatial patterns of genetic diversity shown in Fig. 1bi are spontaneous self-organizing ones; there was no underlying spatial heterogeneity in the environment as a template. These spatial patterns of polymorphism are dynamic because of the motion of hybrid zones and thus are distinct from the static polymorphism found in subdivided and weakly coupled patch models of ecological systems (Karlin & McGregor 1972; Levin 1974).

The genetics of hybrid zones has been studied with increasing attention (Endler 1977; Barton 1979; Barton & Hewitt 1981; Harrison 1986; Barton & Hewitt 1989; Harrison 1993; Nurnberger et al. 1995; Hewitt 2001), and the motion of hybrid zones has been seen in experimental observations (Pearson 2000; Britch et al. 2001; Rohwer et al. 2001). These studies typically analyze the dynamics of hybrid zones based on the difference between the inherent fitness of two competing species where their domains meet. In contrast, our study dealt with the spontaneous emergence of spatial separation between two equally favored types and the motion of hybrid zones by a particular kind of context-dependent fitness difference that arises from spatial distribution and dynamics, not simply from the inherent biological superiority of one genotype. In our model, the local curvature of hybrid zones gave a local context dependence to effective fitness. This may be relevant but has not been considered in studies on hybrid zones in the wild.

Maintenance of Genetic Diversity

We considered the relevance of the dynamics of such heterogeneous spatial patterns in the genotypes (and their associated hybrid zones) to the persistence of genetic diversity and, particularly, the significant role of habitat size and shape. These issues are not typically considered in theoretical or laboratory experimental studies that assume spatially homogeneous populations and that may specifically exclude consideration of edge and boundary effects.

Our results showed that spontaneous pattern formation implies that genetic diversity persists significantly longer in a population with local demes than in a panmictic population when they are subject to disruptive selection (di-

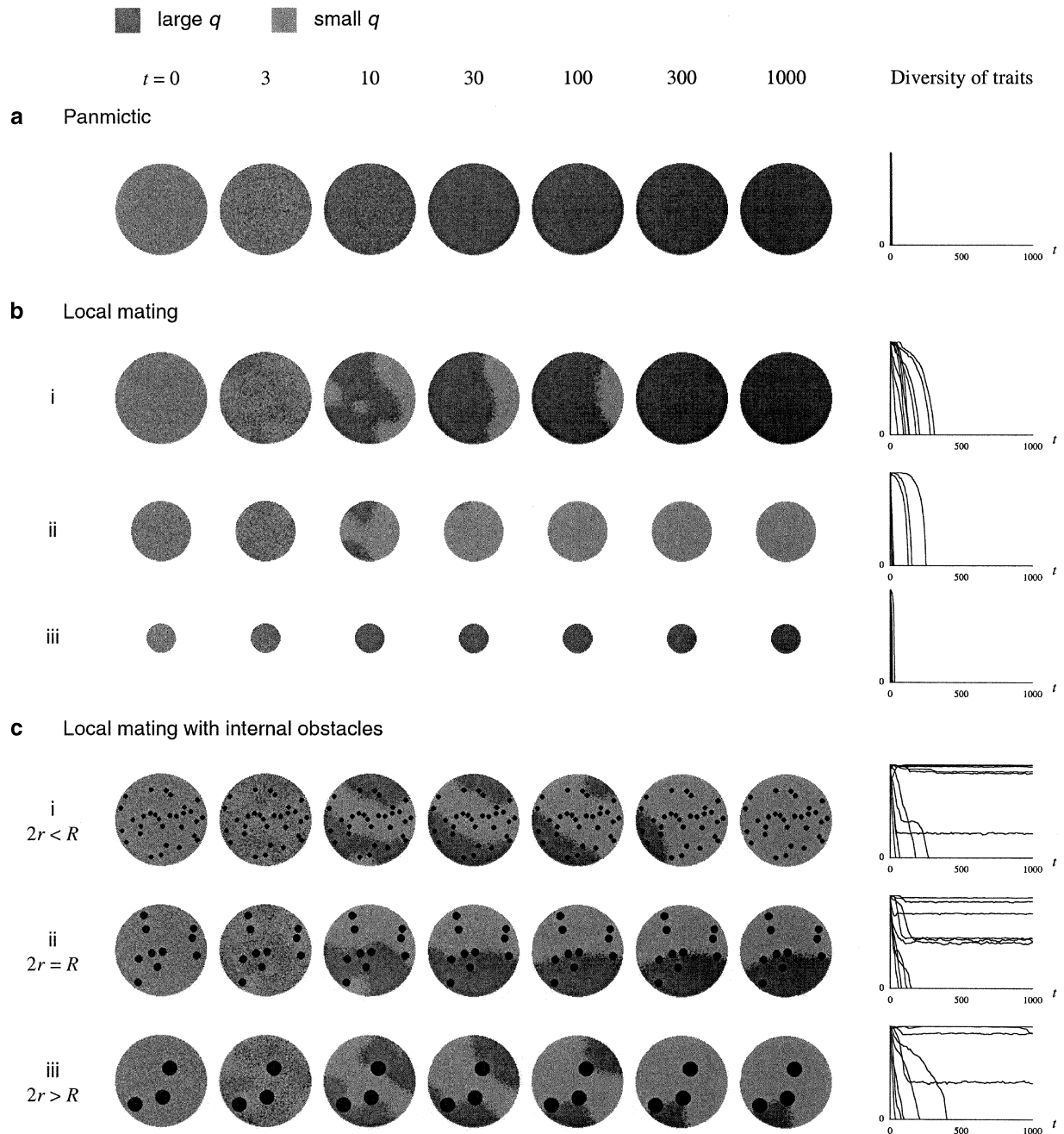


Figure 1. Simulations of pattern formation in a spatially distributed mating population undergoing disruptive selection initiated at $t = 0$. Organisms are characterized by a quantitative trait q (Appendix 1). Large and small q are favored over intermediate values of q by disruptive selection. Seven cases with different conditions are shown. On the right, the time dependence of the diversity is plotted for 10 sample simulation runs. Diversity is characterized by the variance of the trait distribution. The radius of the local demes of organisms, R , is fixed at 10 for all simulations, except in (a) where R is infinite. (a) Simulations of a panmictic population, in which organisms can mate regardless of spatial distance. In this case diversity vanishes only a few generations after the introduction of disruptive selection. (b) Simulations with local mating, in which the size of local demes of organisms is smaller than the size of the habitat. Three cases with different habitat sizes are shown (i, ii, iii). The ratio of the diameter of one local deme to the total habitat diameter is about 16% for i, 25% for ii, and 50% for iii. Local mating causes pattern formation and thus persistence of genetic diversity. The time of persistence is shorter for smaller habitats. (c) Simulations with local mating in a habitat that contains internal obstacles (black spots). Three cases with different obstacle sizes (i, ii, iii) are shown, keeping the total obstacle area almost the same (r is the radius of the obstacles). Habitats with obstacles have a longer persistence of diversity because of the pinning of the motion of hybrid zones at the obstacles. The obstacles with a size $2r = R$ are the most effective (Fig. 2).

versity plots in Fig. 1a & 1b). The time until homogenization increased with the size of the habitat (Fig. 1bi, ii, iii) because the characteristic length scale of spatial patterns showing coarsening behavior grew as a square root of the elapsed time (Sayama et al. 2000) and therefore a longer time was needed to reach homogenization in larger areas. This suggests that protected habitats that are substantially smaller than natural ones will tend to lose genetic diversity more rapidly when disruptive selection occurs. This is of concern because high genetic diversity is a key to the robustness and resilience of biological systems (Tilman & Downing 1994; Holling et al. 1995). The loss of diversity when only one of two original types is left is not only the loss of the other type but also of the many possible crosses between them that might be of higher fitness if environmental changes occur.

Our model is relatively simple; were additional complexity added to further enhance biological realism, the role of emergent pattern formation and boundary effects would most likely become even more profound. As a first step, we examined the effect of adding geographical irregularity on the model organism's distribution pattern.

Effects of Geographical Irregularities

The results shown in Fig. 1b, and those obtained in other studies (Durrett & Levin 1994; Gandhi et al. 1998; Sayama et al. 2000), all apply only to simple habitat structures. To further study the implications of pattern formation, we considered the irregular geographical features of natural and protected habitats. The effects of spatial barriers on static clines have been studied with reaction-diffusion equation models of gene frequency (Slatkin 1973; Nagylaki 1976; Fife 1979). Also, the effects of population subdivision on neutral genetic variation and effective population size have been studied with metapopulation models (Gilpin 1991; Gliddon & Goudet 1994; Hedrick & Gilpin 1997). In contrast with these earlier studies that calculated only stationary solutions (final structures) of genetic variation that reflect reduced migration rates between subpopulations in zero- or one-dimensional spaces, we considered the dynamic movement and transformation of curved hybrid zones and their interactions with localized obstacles in two dimensions.

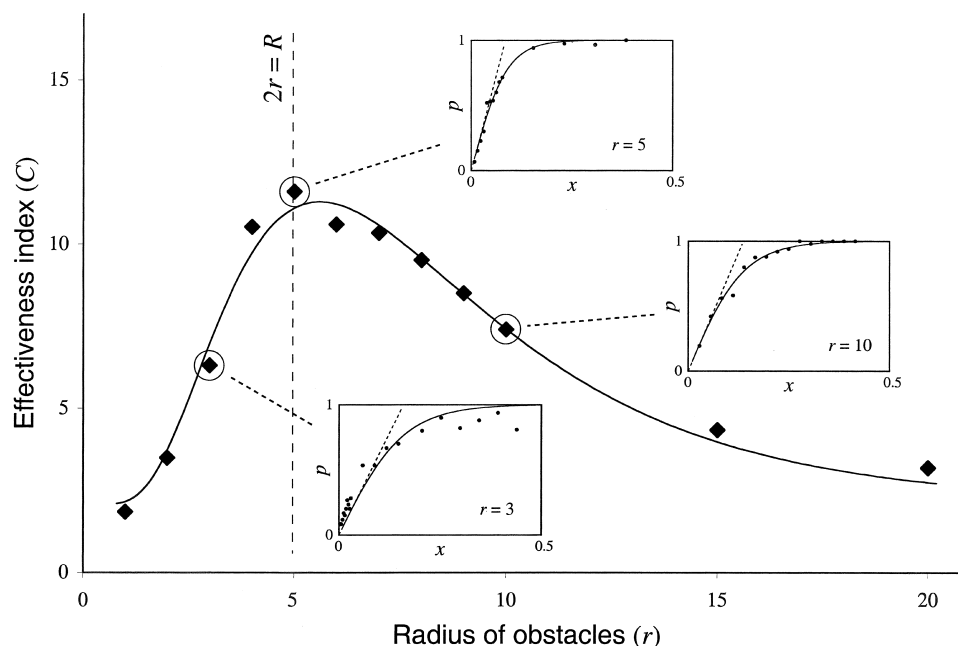


Figure 2. Relationship between obstacle size and effectiveness of obstacles in preserving diversity, as measured by the effectiveness index C as a function of the radius of obstacles r . The effectiveness index is the initial slope of a hyperbolic tangent $p = \tanh(Cx)$ fitted to the simulation results $p(x)$, where p is the probability of maintenance of polymorphism and x is the obstacle fraction (the ratio of obstacle area to total area of the habitat). This measures the efficiency per unit area of the obstacles. A lognormal curve is fitted to the results for reference. This plot has a peak near $2r = R$. Thus, obstacles that are about as large as the radius of local demes are the most effective in maintaining polymorphism (Fig. 1c). The probability of maintenance of polymorphism was measured for each r and x through use of the final outcomes of multiple simulations for 500 generations; thus, without obstacles the result would be zero. Each data point plotted in the embedded $p(x)$ graphs reflects the results of at least 20 runs, with an average of over 50 runs per point. The total number of runs used to obtain these results is over 10,000.

Figure 1c shows examples of simulations representing habitats with internal obstacles of different sizes. Such irregular features inhibited and sometimes eventually stopped the motion of hybrid zones and thus froze coarsening patterns, preventing the reduction of genetic diversity through homogenization by reproductive mixing. We systematically evaluated the effectiveness of the internal obstacles in freezing the coarsening patterns by measuring the probability of persistence of polymorphism. Two parameters were varied: the size of an obstacle and the fraction of obstacle area inside the habitat (the ratio of the area of the obstacles to the total area of the habitat).

We carried out over 10,000 simulation runs to explore the parameter space. Generally, the probability of polymorphism increased as the density of obstacles increased (Fig. 2), due to a decrease in the rate of local patch extinction through the pinning of hybrid-zone motion. This effect saturates at high obstacle densities. In order to compare obstacles of different radius, we defined *effectiveness index*, which measures the efficiency per unit area of internal obstacles in maintaining diversity, as the slope of a hyperbolic tangent fitted to $p(x)$, where p is the probability of maintenance of polymorphism and x is the obstacle fraction. Significantly, the effectiveness index was maximal when the size of obstacles was equal to the radius of a local deme.

This can be understood in that obstacles smaller than the radius of a deme are not large enough to hamper local reproductive mixing and thus do not effectively freeze coarsening patterns. Obstacles at the size of the radius of a deme can freeze the patterns. Obstacles larger than this are not much more effective, but they use a larger area and thus are less efficient. This result weakly depended on the circular shape of the obstacles we used. It is expected that if one used star-shaped obstacles, polymorphism would be retained, with smaller covered areas having larger perimeters. The specific scale (diameter) of the most effective obstacles should be independent of their shape, however, because it corresponds to the range of the mating processes.

Discussion

Our results on the implications of spatial pattern formation for genetic diversity in natural and artificial habitats can be summarized as follows. In the reference case, when selection occurred in the context of complete reproductive mixing of the population, population diversity declined rapidly. Local mating resulted in the preservation of trait variation and genetic diversity over substantially longer times according to our spatially extended population model. Even in the context of local mating, however, diversity eventually declined and did

so more rapidly for smaller habitats. Local geographical irregularities can further extend the retention of higher diversity. The effect of localized irregularities should be contrasted with the effect of habitat partitioning. Absolute partitioning is not desirable because it will result in small populations with a high probability of extinction. Spatial obstacles, even though they do not partition the space, inhibit complete reproductive mixing and are effective in maintaining genetic diversity in a geographically contiguous population. Our results imply that where it is possible to design such obstacles, using obstacles whose size approximates the radius of the local demes may lead to maintenance of polymorphism with minimal sacrificed habitat area. Other factors should also be considered in practical planning, such as the total availability of suitable habitats and the potential influences of obstacle construction.

In additional studies we found that convoluted habitat boundaries also froze coarsening patterns. The implications of this result are counterintuitive because it implies that genetic diversity could be enhanced by increasing the ratio of habitat edge to area. For actual habitat design, however, taking advantage of this effect would also increase undesirable edge effects, such as deterioration of microclimate or invasion by predator and weed species. Thus, from a practical viewpoint, the external boundaries of protected habitats should probably be kept as short as possible because they are generally sources of environmental degradation and encroachment.

Our results suggest that internal obstacles in habitats might be introduced to enhance the maintenance of genetic diversity, but what would these obstacles actually be in any practical application? In an aquatic system they could be islands of a different physical environment, such as rocky reef on a mud or sand bottom. In a terrestrial park they could be fenced-off areas that are protected both inside and outside and in which, for example, conservation and ecology management experiments could be conducted. In an era when the trend in zoological parks is to become as natural as possible, it might seem abhorrent to unleash interior designers upon the small nature reserves that we fondly but erroneously regard as accessible mini-wildernesses (e.g., Nairobi National Park). With few vast areas of true wilderness remaining in the world, current strategies for preservation of endangered species call for the maintenance of as many independent, self-sustaining populations as possible to maximize both total population size and genetic diversity. Our work suggests that, at least for some species, intelligent design can maximize the conservation value of any one of these highly constrained remnant habitats.

Finally, our results demonstrate the existence of powerful linkages between biological processes at three dif-

ferent scales. Spontaneous pattern formation and its interplay with geographical features are emergent traits of ecological populations, caused by attributes of the organismal reproductive process that reflect genetic recombination inside each organism. This is an important example that contradicts the erroneous but commonly held assumption that it is possible to neatly partition evolutionary effects at different spatial scales, to study a molecule, individual, population, metapopulation, species, or ecosystem. Indeed, when hierarchical structure is considered a signal attribute of complex systems, it is the dynamic linkages among the levels, rather than the number of levels themselves, that should probably be the focus of attention. We emphasize that the self-organization of dynamic patterns through multiscale interactions is a characteristic property of diverse complex systems, including spatially distributed living systems (Bar-Yam 1997), and that recognition of this property can lead to deeper insights into community organization and evolution.

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Appendix 1. Details of the individual-based computer simulation model of spatial patterns of genetic diversity.

The model used for this research is an individual-based model with a two-dimensional spatial grid of sites in which each site can contain one mature monoecious organism with an n -locus genome. Haploid and diploid models both yield the same results, and we present the results obtained with haploid models. There are m possible alleles for each locus. The genome of an organism determines a quantitative trait q . Each allele is assumed to independently and linearly contribute to the trait (i.e., $q = \sum a_i$, where a_i is an integer ranging from 0 to $m - 1$ that represents the effect of the allele at the i th locus). The possible traits range from 0 to $n(m - 1)$. Modifying the assumption of linearity of allelic effects to traits can change the relative allelic diversity between different loci, but does not significantly change the general results on the dynamics of spatial patterns we have reported here. Fitness is a function of the quantitative trait, $f(q)$, which is lowest for intermediate values of the trait and highest for the two extreme values. We have tested several different fitness assignments, including V-shaped, quadratic, and discrete (stepwise) dependences of the fitness as a function of the trait. All these assignments yielded the same results. Reproduction in the model assumed random genetic crosses between two parent organisms (i.e., the allele at each locus is inherited independently from either of the two parents). The parent organisms were chosen randomly out of a local neighborhood of radius R around each site. Such mating processes were repeated P times per site per generation, then the fittest individual of the P offspring at each site became the selected parent at the next generation. For the sake of simplicity, we did not consider the effects of mutation.

The parameters used in the figures were $n = 10$, $m = 10$, $P = 10$, $R = 10$, and the fitness of a trait was a V-shaped curve (i.e., the fitness was linearly related to the distance of the trait value from the median possible value [$f(q) = |q - n(m - 1)/2|$]). The habitat was a circular space with a diameter of 127 site widths (except for Fig. 1, Bii, iii). For simulations described in Fig. 1c and Fig. 2, circular obstacle patterns on which organisms cannot exist are randomly placed inside the habitat in such a way that they neither overlap one another nor stick out from the habitat. The initial conditions consisted of organisms with randomly assigned genotypes, a situation that can arise when selection becomes disruptive after being nondisruptive.

